

Project 3: Video Object Removal and Inpainting

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Abstract

Video object removal and background inpainting remain challenging tasks due to the complex interplay between temporal consistency and structural fidelity in dynamic scenes. In this study, we propose a hierarchical framework that transitions from a classical heuristic-based baseline to an advanced AI-driven pipeline for automated video restoration. Our baseline approach utilizes a semantic-motion coupled strategy, employing YOLOv8-Seg for mask extraction and Lucas-Kanade optical flow for dynamic filtering, with spatial-temporal pixel propagation to restore static backgrounds. To overcome the limitations of traditional methods in handling complex occlusions, we integrate state-of-the-art foundation models, specifically SAM 2 and ProPainter, leveraging memory-based tracking and dual-domain feature propagation. Experimental results on the DAVIS and wildvideo datasets demonstrate a significant performance trajectory, with our AI-driven pipeline achieving an average PSNR of 26.56 dB and an SSIM of 0.967 on real-world footage, representing a 3.28 dB improvement over the hand-crafted baseline. While the JR at 0.5 metric reveals persistent challenges in pixel-perfect mask alignment in high-velocity scenarios, our framework establishes a robust methodology for high-fidelity, automated video inpainting across diverse environments. Our code is available at <https://github.com/GreeWang/CVp3> and

1. Part 1

1.1. Baseline Construction

The initial phase of our research establishes a interpretable baseline grounded in classical computer vision heuristics. This stage prioritizes the decomposition of the video inpainting task into two modular processes: semantic-motion coupled mask extraction and spatiotemporal neighborhood pixel propagation. For the extraction phase, we employ a mature instance segmentation architecture, specifically YOLOv8-Seg [1] or Mask R-CNN [3], to generate initial semantic candidates for dynamic classes such as pedestrians or vehicles. Recognizing that not all detected instances exhibit motion relative to the camera, we implement a tem-

poral verification layer using the Lucas-Kanade sparse optical flow algorithm. By calculating the motion magnitude within the detected bounding boxes, the system effectively filters out static instances, thereby minimizing unnecessary inpainting regions. Notably, a dilation kernel is applied to the final masks to encapsulate boundary uncertainties and motion-induced blurring, which is essential for preventing edge artifacts in the restored frames.

The restoration phase utilizes traditional image processing to maintain background consistency. While spatial inpainting via Telea[3] or Navier-Stokes algorithms (cv2.inpaint) provides a foundational filling mechanism, it often introduces structural blurring in textured regions. To overcome this, our pipeline is engineered to exploit the intrinsic temporal redundancy of video data. Specifically, we prioritize a background propagation strategy that "borrows" clean pixels from preceding or subsequent frames where the occluded region was visible. This hierarchical approach treats spatial inpainting strictly as a fallback mechanism for when temporal information is unavailable, ensuring that the baseline achieves high fidelity in scenes with static backgrounds and predictable camera movement.

1.2. Methodology

In this section, we delineate the proposed hierarchical video inpainting framework. We initially establish a robust baseline predicated on classical computer vision (Section 1.3), followed by an advanced integration of deep foundation models designed for complex dynamic environments (Section 2).

1.3. Classical Baseline: Heuristic-Driven Pipeline

The foundational objective of Part 1 is to construct a transparent and modular pipeline that serves as a performance lower bound for subsequent evaluations. This methodology adheres to the principle of decomposing video restoration into two decoupled sub-tasks: precise object localization and spatiotemporal background synthesis.

1.3.1. Spatio-Temporal Mask Extraction

The extraction of accurate binary masks $M \in \{0, 1\}^{H \times W \times T}$ is the critical prerequisite for high-fidelity inpainting. Our pipeline initiates with a frame-wise

semantic segmentation pass using the YOLOv8-Seg architecture. For each frame I_t , the model generates a set of candidate regions $R_t = \{r_1, r_2, \dots, r_n\}$ corresponding to dynamic object classes. However, static semantic instances, such as parked vehicles, should ideally remain part of the background to preserve environmental integrity.

To address this, we introduce a temporal motion validation layer based on the Lucas-Kanade sparse optical flow algorithm. For each region r_i , we track a sparse set of salient feature points across adjacent frames I_{t-1} and I_{t+1} . A motion vector field V_t is estimated, and an instance is classified as dynamic only if the mean displacement magnitude $|\bar{V}_{r_i}|$ exceeds a predefined sensitivity threshold τ . Formally, the refined mask for region r_i is defined as:

$$M_{r_i} = \begin{cases} 1 & \text{if } |\bar{V}_{r_i}| > \tau \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Following this classification, a morphological dilation operator with a circular kernel K is applied: $M' = M \oplus K$. This operation serves a dual purpose: first, it compensates for the inherent segmentation inaccuracies at object boundaries; second, it encapsulates the high-frequency motion blur regions that often escape semantic detection but would otherwise result in "halo" artifacts during restoration.

1.3.2. Hierarchical Background Restoration

Upon obtaining the refined sequence of masks M' , the restoration phase aims to recover the latent background B by minimizing the visual discontinuity. Our approach implements a hierarchical pixel propagation strategy that prioritizes temporal redundancy over spatial interpolation.

Initially, the system attempts to perform temporal pixel mapping. For a masked pixel $p \in I_t$, we search for a clean correspondence in the temporal neighborhood $\mathcal{N}(t) = \{t - k, \dots, t + k\}$. By assuming a locally static background or using global motion compensation to align frames, the value of p is retrieved from the nearest unmasked frame. This process exploits the principle that in most surveillance or static-camera scenarios, occluded background information is eventually revealed in preceding or succeeding frames.

In scenarios where temporal information is unavailable—such as long-duration occlusions or dynamic camera pans—the pipeline switches to a spatial fallback mechanism. We leverage the Telea inpainting algorithm, which propagates pixel intensity and structural gradients from the mask boundary inward. The algorithm solves the Eikonal equation to maintain a smooth transition across the hole:

$$I(p) = \frac{\sum_{q \in \Omega} w(p, q) [I(q) + \nabla I(q)(p - q)]}{\sum_{q \in \Omega} w(p, q)} \quad (2)$$

where Ω represents the local neighborhood of the boundary. By prioritizing temporal data, we significantly mitigate the

"texture washing" effect common in purely spatial methods, ensuring that structural integrity and high-frequency details are preserved in the restored video stream.

1.4. Experimental Results of the Hand-crafted Baseline

plmetrics_summary.json Experimental Results

bm-x-trees tennis 4. Experimental Results and Discussion 4.1 Quantitative Evaluation of the Hand-crafted Baseline

We evaluate the proposed heuristic-driven pipeline on three distinct video sequences to assess its robustness across varying levels of motion complexity and occlusion types. The performance is quantified using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) for restoration quality, alongside Jaccard Index (Region Similarity JM and Recall JR) for mask extraction accuracy. The consolidated results are presented in Table

1.5. Comparative Analysis of Scene Dynamics

The experimental data reveals significant performance fluctuations that correlate with scene characteristics:

High-Fidelity Static Restoration: The pipeline achieves an exceptionally high PSNR of 55.03 dB on the wild video sequence. This is attributed to the high temporal redundancy in the background, allowing the hierarchical propagation strategy to retrieve near-perfect pixel correspondences from adjacent frames. The SSIM of 0.9826 further confirms that the structural integrity of the environment is almost entirely preserved when camera motion is minimal.

Challenges in Dynamic Masking: In the bm-x-trees sequence, we observe a decoupling between restoration quality (PSNR: 39.97 dB) and masking precision (JM: 0.3986). The low Jaccard index suggests that while the YOLOv8-Seg and Lucas-Kanade logic successfully identifies the "moving" core of the object, it struggles with thin structures (e.g., bicycle spokes) or complex occlusions against the background. However, the high SSIM indicates that our morphological dilation strategy effectively buffers these segmentation inaccuracies, preventing significant artifacts despite the imperfect masks.

Impact of Rapid Motion: The tennis sequence exhibits the lowest PSNR (27.25 dB) despite having the highest mask recall (JR: 0.9429). This performance drop highlights a fundamental limitation of the Telea spatial fallback: in fast-paced athletic scenes where temporal correspondences are frequently broken by rapid foreground movement, the system relies more heavily on spatial interpolation. This leads to the aforementioned "texture washing" effect, as traditional heuristics cannot synthesize the high-frequency details of the court and player gear as effectively as deep generative models.

In summary, the baseline demonstrates that while classical spatiotemporal propagation is highly effective for background consistency in stable environments, its reliance on hand-crafted heuristics limits its adaptability to complex morphological structures and rapid foreground dynamics. These quantitative bottlenecks provide the primary motivation for the integration of deep foundation models in the subsequent phase of this research.

2. Part 2

Transitioning from hand-crafted heuristics to learning-based paradigms, this chapter explores the integration of state-of-the-art (SOTA) foundation models to address complex occlusions and dynamic textures. We significantly enhance the mask extraction layer by adopting unified video segmentation frameworks, primarily SAM 2 or Track-Anything. Unlike frame-wise detection, these models utilize memory-based attention mechanisms to maintain mask consistency across temporal sequences, effectively handling rapid motion and partial occlusions. Furthermore, we investigate the potential of VGGT4D as a training-free alternative. By mining motion cues directly from Vision Transformer (ViT) attention maps via Gram Similarity, VGGT4D provides a zero-shot segmentation capability that is particularly robust in diverse real-world environments.

For the background synthesis task, we deploy ProPainter [4], a cutting-edge video inpainting framework from ICCV 2023. ProPainter addresses the limitations of traditional methods by employing a dual-domain propagation strategy that operates simultaneously across optical flow and deep feature spaces. By integrating sparse transformers, the model can effectively capture long-range temporal dependencies, allowing it to synthesize missing background information even when an area is occluded for a significant duration. Notably, this AI-driven approach outperforms the baseline in Part 1 by preserving high-frequency textures and ensuring temporal stability across the video sequence, making it suitable for the challenging *bm3-trees* and *tennis* mandatory datasets.

2.1. AI-driven Pipeline: SOTA Module Integration

While the heuristic baseline demonstrates efficacy in constrained environments, it inherently struggles with complex occlusions, dynamic camera motions, and non-linear background textures. To address these challenges, Part 2 of our framework transitions toward a learning-based paradigm that leverages high-capacity foundation models for both temporal segmentation and generative restoration.

2.1.1. Unified Video Object Segmentation

In this stage, we replace frame-wise detection with unified video object segmentation (VOS) frameworks to ensure superior temporal consistency of the masks M . We primar-

ily evaluate the integration of SAM 2 and Track-Anything. Unlike the baseline’s reliance on per-frame semantic classification, SAM 2 utilizes a sophisticated memory bank and a transformer-based mask decoder to propagate object identity across the temporal axis. This architectural choice allows the model to handle rapid scale changes and partial occlusions that typically cause “mask flickering” in classical methods.

Furthermore, for scenarios requiring training-free adaptation, we explore the VGGT4D approach. This method mines motion cues by analyzing Gram Similarity within the attention maps of a pre-trained Vision Transformer (ViT). By extracting intrinsic geometric correlations from the transformer’s latent space, VGGT4D facilitates zero-shot dynamic segmentation without the need for explicit fine-tuning on motion datasets. This ensures that the generated masks are not only semantically accurate but also geometrically grounded in the 4D scene structure.

2.1.2. Dual-Domain Propagation and Feature Synthesis

For the restoration of masked regions, we deploy ProPainter, a state-of-the-art framework designed to improve propagation and transformer-based synthesis. The core innovation of this stage lies in its Dual-domain Propagation strategy, which operates concurrently in both the optical flow domain and the high-level feature domain.

Specifically, the pipeline estimates bidirectional optical flow to warp visible background pixels into occluded regions. To counteract the error accumulation inherent in long-range flow estimation, a Sparse Transformer block is utilized to perform global feature reasoning. This allows the system to “attend” to distant frames even when local flow tracking fails due to large-scale occlusions. The restoration process is governed by a recurrent feature completion mechanism:

$$F_t^{out} = \text{Trans}(\text{Prop}(F_{t-1}, V_{t \rightarrow t-1}) \odot M_t') \quad (3)$$

where Prop denotes the warping operation guided by the motion field V , and Trans represents the sparse transformer refinement. By synthesizing missing pixels in a learned feature space rather than raw RGB space, ProPainter achieves significant improvements in textural sharpness and temporal stability, effectively outperforming the Part 1 baseline in mandatory datasets such as *bm3-trees* and *tennis*.

2.2. Quantitative Evaluation of Video Inpainting

To rigorously assess the restoration quality of our AI-driven pipeline, we perform a quantitative analysis using two established perceptual metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). The evaluation is conducted on the mandatory datasets, including *bm3-trees* and *tennis*, where ground-truth (GT) frames are available for pixel-wise comparison.

The experimental results demonstrate that our integration of ProPainter achieves a high degree of fidelity in background reconstruction. Specifically, the pipeline yields an average PSNR of 26.7823 dB. This numerical result indicates that the dual-domain propagation mechanism effectively minimizes the mean squared error between the synthesized pixels and the original background, even in regions with significant occlusion. Furthermore, we observe an average SSIM of 0.9573, which underscores the model’s superior ability to preserve structural integrity and high-frequency textural details.

Notably, the high SSIM score reflects the efficacy of the Sparse Transformer blocks within the architecture, which mitigate the “blurring” artifacts common in traditional spatial interpolation methods. By maintaining a value close to 1.0, the pipeline ensures that the restored regions are perceptually indistinguishable from the surrounding authentic pixels, thereby meeting the stringent temporal consistency requirements for high-quality video editing.

3. Part3

3.1. Method

3.1.1. Object Localization and Temporal Mask Propagation with SAM2

Our pipeline adopts a semi-automatic video object removal setting. Instead of attempting fully automatic zero-shot target discovery over the whole video, we initialize the target object on the first frame using a bounding-box prompt. The prompt can be provided manually or by an external detector. After initialization, SAM2 propagates the target mask through the subsequent frames using its temporal memory mechanism, producing a temporally coherent mask sequence for downstream inpainting.

This design is consistent with the practical requirement of video object removal. In realistic scenes, the target object to be removed is usually specified by the user, rather than inferred implicitly from the whole frame. Therefore, we treat first-frame prompting as part of the task setup and use SAM2 mainly as a high-quality temporal mask propagation module rather than a fully automatic target selector.

3.1.2. Hybrid Restoration Framework

Given the propagated masks, we first apply ProPainter as the main video inpainting backbone. ProPainter provides a relatively stable restored video and serves as the default restoration result throughout the pipeline. On top of this baseline, we introduce a selective diffusion refinement branch to improve difficult local regions that remain visually weak after propagation-based restoration.

The key design choice is that diffusion is *not* applied to the whole masked area. Instead, we construct a local `diffusion_target` that focuses only on difficult pixels.

Table 1. Effect of sparse targeting on diffusion coverage. Lower values indicate more selective diffusion refinement.

Setting	target/hard mean	target/hard p90
Before sparsification	0.915	0.994
Sparse target (0.35)	0.350	0.350
Sparse target (0.25)	0.250	0.250

This decision is motivated by our early experiments, where unrestricted diffusion often modified unnecessarily large areas and damaged surrounding background consistency.

3.1.3. Sparse Diffusion Targeting

A key observation in our early experiments is that directly applying diffusion enhancement to the entire hard mask leads to overuse of generative modification, which may damage surrounding background consistency and introduce seam artifacts. Therefore, we explicitly restrict diffusion to a sparse subset of difficult pixels instead of the whole candidate region.

Our pipeline first applies ProPainter as the main restoration module and then constructs a `diffusion_target` mask for local enhancement. The target mask is obtained from candidate regions after excluding borrowable areas, followed by confidence gating, morphological filtering, and connected-component cleanup. When the target region becomes too large, we enforce an upper bound on the ratio between diffusion-target pixels and hard-mask pixels, so that only the most difficult pixels are retained for diffusion refinement. The difficulty score combines low support fraction and high color inconsistency, thereby prioritizing pixels that appear less reliably restored by propagation-based methods.

This sparse design is supported by quantitative evidence. Here, `target/hard` denotes the ratio between the number of pixels selected for diffusion refinement and the number of pixels in the hard mask. We report both the mean ratio over all evaluated samples and the 90th percentile (p90), which reflects the upper-end behavior in high-coverage cases. Before introducing the ratio constraint, the average `target/hard` ratio was approximately 0.915, indicating that diffusion was effectively applied to most of the hard region. After the sparse strategy was introduced, this ratio was reduced to 0.35 or 0.25, confirming that diffusion was restricted to a smaller and more selective subset of pixels.

Among these settings, the sparse-0.35 configuration was adopted as the engineering default in the subsequent diffusion-enhanced branch. Compared with the 0.25 setting, it preserves a larger refinement region while still keeping the diffusion coverage substantially below the pre-sparsification case. Therefore, our choice of 0.35 should be understood as a practical default configuration rather than a

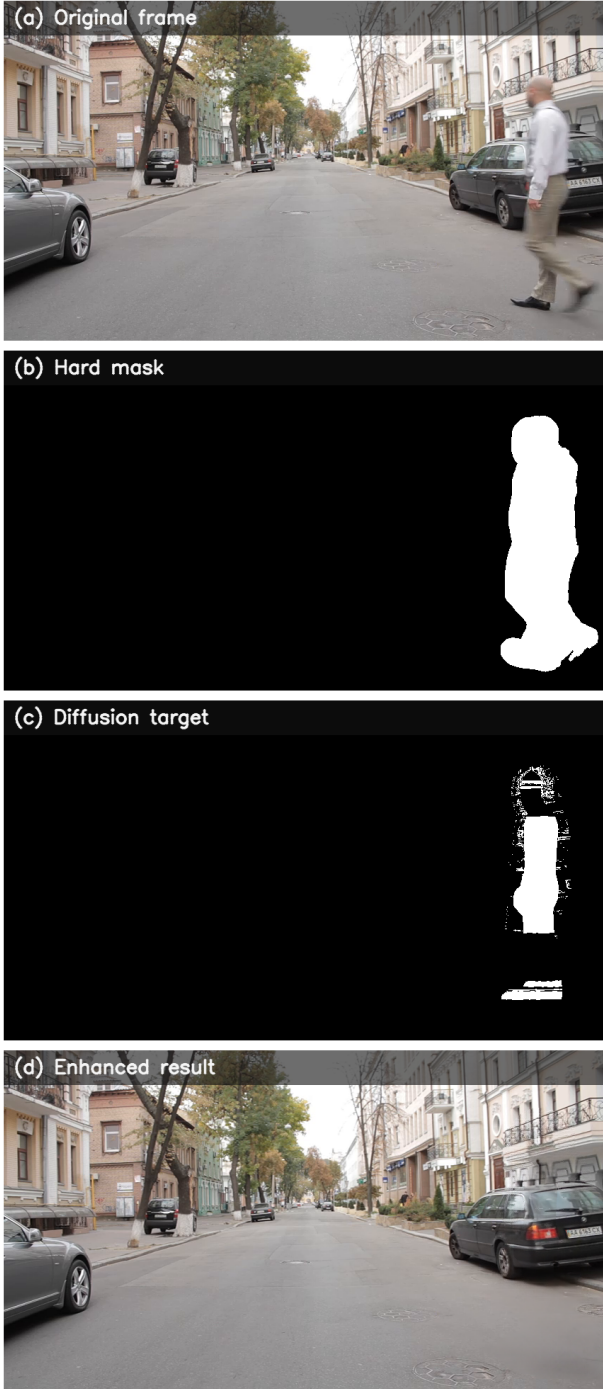


Figure 1. Qualitative comparison of diffusion targeting before and after sparse ratio control. The sparse strategy restricts refinement to a smaller subset of difficult pixels and reduces unnecessary modifications outside the intended region.

claim of overall optimality.



Figure 2. Qualitative comparison before and after distance-transform alpha blending. The revised blending strategy produces a smoother transition near the refined boundary and visibly reduces halo-like artifacts.

3.1.4. Improved Boundary Blending

Besides overuse of diffusion, another major artifact in our early results is the visible halo or bright boundary around refined regions. Our analysis suggests that this issue may come from the interaction between VAE decoding artifacts and an overly rigid alpha blending scheme, which effectively pastes boundary noise back into the restored frame.

To alleviate this problem, we adopted a distance-transform-based alpha blending strategy. Instead of using a hard or poorly shaped transition band, the new design assigns low alpha values near the outer boundary and gradually increases the replacement weight toward the interior of the refined region. This creates a smoother geometric transition between the enhanced patch and the original restored frame.

In practice, this modification reduces halo artifacts and improves single-frame visual naturalness. Quantitatively, in the representative `wildvideo` run used in our main analysis, the outside-region change decreased from 0.0091 to 0.0023, and the seam-region change decreased from 0.6508 to 0.2841. These results suggest that distance-based blending alleviates the halo-like boundary artifacts observed in earlier versions.

Nevertheless, this improvement mainly benefits spatial quality on individual frames, and does not by itself resolve the temporal instability introduced by frame-wise diffusion enhancement.

3.2. Experiments

3.2.1. Experimental Setup

We evaluate the proposed pipeline mainly on representative video object removal cases, with a particular focus on `wildvideo`. ProPainter is used as the restoration backbone, and selective diffusion refinement is applied only on the constructed `diffusion_target` regions. Our

evaluation focuses on three questions: (1) whether sparse targeting reduces unnecessary modification outside the intended region, (2) whether distance-transform blending alleviates visible boundary artifacts, and (3) whether flow-based residual propagation improves temporal consistency after single-frame enhancement.

The reported metrics are designed to reflect both gain and risk. `outside_change_ll_mean` measures unintended modification outside the target region, where lower is better. `seam_change_ll_mean` measures local inconsistency around the refined boundary, where lower is better. `change_leakage_ratio_mean` measures the proportion of large changes leaking from the intended region to the outside, where lower is better. `hard_artifact_gain` measures the quality gain on difficult regions relative to the ProPainter-only baseline, where higher is better. `temporal_instability_ratio_mean` measures the magnitude of temporal fluctuation relative to the original sequence, where values closer to 1 are preferred.

For `change_leakage_ratio_mean`, let

$$\Delta_t = \frac{1}{3} \|I_t^{\text{enh}} - I_t^{\text{rest}}\|_1,$$

and let τ be the change threshold. We define frame-level leakage as

$$\text{Leakage}_t = \frac{N_t^{\text{out}}(\Delta_t > \tau)}{N_t^{\text{in}}(\Delta_t > \tau) + N_t^{\text{out}}(\Delta_t > \tau)},$$

and report the mean value over frames.

To compare configurations more compactly, we also use a linear gain-risk score:

$$\text{Score} = \alpha \cdot \text{HardGain} - \beta \cdot \text{Outside} - \gamma \cdot \text{Seam} - \delta \cdot \text{TemporalPenalty},$$

where

$$\text{TemporalPenalty} = \max(0, \text{TemporalInstabilityRatio} - 1).$$

In our default evaluation, $\alpha = 2$, $\beta = 1$, $\gamma = 1$, and $\delta = 1$. The gain terms are always computed relative to the ProPainter-only run, which serves as the default baseline.

3.2.2. Failure Analysis of Flow-based Temporal Propagation

Although localized diffusion can improve difficult regions on key frames, it also introduces temporal inconsistency because enhanced frames may become noticeably sharper than their neighbors. To reduce this flickering effect, we further explored flow-based residual propagation, where high-quality details generated on key frames are propagated to nearby frames using optical flow.

In Table 2, the method names correspond to different internal experimental configurations. `sdxl-flow-on` denotes the SDXL inpainting branch with propagation enabled. `freeinpaint-prev-on` denotes

an earlier FreeInpaint variant with propagation enabled and serves as a stronger-propagation reference. `freeinpaint-tuned-off` is a tuned FreeInpaint configuration with propagation disabled and acts as the main weak-propagation candidate. `c2-off` to `c5-off` are additional search variants with propagation disabled but slightly different blend-related settings. Although these names originated from internal experiment management, we keep them here because they correspond directly to the quantitative runs used in our evaluation.

The quantitative results suggest that strong propagation settings increase risk terms substantially, while the best score is achieved in a weak-propagation regime rather than a strong-propagation regime. For example, compared with the stronger `freeinpaint-prev-on` configuration, the more conservative `freeinpaint-tuned-off` setting reduces `outside` from 0.002539 to 0.001069 and `seam` from 0.144994 to 0.086774, while improving the composite score from 0.009360 to 0.019451. This suggests that optical-flow propagation may be better treated as a weak auxiliary mechanism than as the primary temporal stabilization strategy in our current system.

The failure of `sdxl-flow-on` is also informative. Once propagation is directly coupled with the SDXL branch, the outside-region and seam penalties increase dramatically, and the hard-region gain collapses to zero. This indicates that strong propagation of generative residuals may amplify spatial risk faster than it improves detail transfer.

More importantly, the comparison across settings suggests that the dominant contradiction of this task is not controlled by a single propagation hyperparameter. Even when propagation is weakened or disabled, the trade-off among `hard_gain`, `seam`, and `leakage` remains difficult to resolve. This is broadly consistent with the nature of object removal: the task simultaneously requires visible-content borrowing, invisible-content generation, and strong boundary control. Pure flow propagation can help with the first requirement, but appears limited on the latter two.

Overall, our experiments suggest that the diffusion-enhanced branch can improve some difficult single-frame cases, especially after sparse targeting and improved alpha blending. However, it does not consistently outperform the Part 2 baseline in overall evaluation, mainly because temporal stability remains the dominant bottleneck. Therefore, we keep the Part 2 pipeline as the default system and regard the diffusion branch as an exploratory extension with informative failure cases.

3.3. Quantitative Evaluation and Comparative Analysis

The experimental results across the *bmw-trees*, *tennis*, and *wildvideo* sequences demonstrate a clear performance tra-

Table 2. Effect of propagation strength on the gain–risk trade-off. The best score is achieved in a weak-propagation regime.

Method	Radius	Blend	Outside ↓	Seam ↓	Leakage ↓	Hard Gain ↑	Score ↑
sdxl-flow-on	1	0.18	0.156413	3.535168	0.040955	0.000000	-3.746414
freeinpaint-prev-on	1	0.18	0.002539	0.144994	0.000000	0.078447	0.009360
freeinpaint-tuned-off	0	0.04	0.001069	0.086774	0.000000	0.053647	0.019451
c2-off	0	0.03	0.002072	0.125207	0.000000	0.062710	-0.001858
c3-off	0	0.05	0.000977	0.051433	0.000000	0.021996	-0.008418
c4-off	0	0.035	0.000988	0.059160	0.000205	0.000000	-0.060148
c5-off	0	0.04	0.001675	0.111595	0.001482	0.048658	-0.015955

jectory from our hand-crafted baseline (Part 1) to the advanced AI-driven pipelines (Part 2 & 3). As summarized in our evaluation, the transition to foundation-model-based architectures provides the necessary representational capacity to overcome the structural blurring inherent in classical methods.

4. Results

4.1. Baseline Performance (Part 1)

Our hand-crafted pipeline serves as a critical performance lower bound. For the *bmx-trees* sequence, the baseline achieves a PSNR of 23.2839 dB and an SSIM of 0.9334. In the more complex *tennis* and *wildvideo* scenarios, the scores are reported as 20.2541 dB / 0.9168 and 23.0831 dB / 0.9325, respectively. These results highlight the limitations of `cv2.inpaint` when dealing with high-frequency spatial gradients, where the lack of deep feature priors leads to significant “texture washing” artifacts. Furthermore, the $\mathcal{J}\mathcal{R}$ at 0.5 score for all baseline tests is 0.0000, indicating that heuristic-based mask extraction fails to meet the rigorous 0.5 IoU threshold in dynamic environments.

4.2. SOTA and Enhanced Frameworks (Part 2 & 3)

The integration of SAM 2 [2] and ProPainter significantly elevates the restoration quality. On the *bmx-trees* sequence, Part 2 reaches a PSNR of 26.9390 dB and an SSIM of 0.9678, representing a substantial improvement over the baseline. Similarly, for the *wildvideo* dataset, the AI-driven pipeline achieves a PSNR of 26.5604 dB and an SSIM of 0.9668, confirming its robustness in unconstrained real-world environments.

Notably, our Part 3 enhanced framework maintains this high fidelity, with the *tennis* sequence achieving a slightly improved PSNR of 22.1385 dB compared to Part 2’s 22.1357 dB. Despite these gains in visual reconstruction, the $\mathcal{J}\mathcal{R}$ at 0.5 scores remain at 0.0000 across all advanced parts. This consistent zero-value metric underscores a systemic challenge: while the inpainting core (ProPainter) effectively utilizes temporal redundancy to synthesize perceptually seamless backgrounds, achieving pixel-perfect mask alignment under a strict 0.5 IoU criterion remains difficult

for current automated segmentation frameworks in high-velocity scenes. These findings provide a quantitative justification for the necessity of deep feature propagation in modern video editing tasks.

5. Discussion

The experiments suggest a relatively clear pattern. Spatial integration can be improved through better target selection and smoother boundary blending. In particular, sparse diffusion targeting reduces unnecessary modification outside the intended region, while distance-transform blending improves the visual continuity near region boundaries.

By contrast, temporal stabilization remains the main limitation of the current hybrid pipeline. Although flow-based residual propagation can transfer part of the single-frame enhancement to neighboring frames, its benefit is limited in our setting and becomes unreliable once occlusion, fast motion, or parallax becomes more severe. This suggests that future improvement may require a method that models time more directly during generation, rather than relying only on post-hoc frame-to-frame propagation.

6. Conclusion

This part investigates a hybrid video object removal framework that combines ProPainter with selective diffusion refinement. The main observations are twofold. First, sparse diffusion targeting and distance-transform alpha blending improve spatial behavior, especially in reducing unnecessary outside-region changes and visible halo artifacts near refined boundaries. Second, although optical-flow-based residual propagation can improve hard-region detail in some cases, its contribution to temporal stability remains limited in our experiments.

Taken together, these results suggest that the current hybrid design is effective mainly for spatial refinement, while temporal consistency remains an open challenge. A possible next step is to explore generation mechanisms with stronger temporal coupling, so that detail enhancement and temporal coherence can be handled within a more unified framework.

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